

DAY 4 - PLACEMENT

21 May 2026 09:06 AM

FORM:

```
import React, { useState } from 'react'
function UseForm() {
  const [formData, setFormData] = useState({
    firstName: "",
    lastName: "",
    email: "",
    dob: ""
  })
  const [submittedData, setSubmittedData] = useState(null)
  const [isSubmitted, setIsSubmitted] = useState(false)
  const handleSubmit = (e) => {
    e.preventDefault();
    setSubmittedData(formData);
    setIsSubmitted(true);
  }
  return (
    <div style={{ padding: "20px", width: "400px", margin: "0 auto" }}>
      <h1>Form</h1>
      { !isSubmitted ? <p>Please fill out the form and submit.</p> : <p>Thank you for
submitting the form!</p> }
      <form onSubmit={handleSubmit}>
        <label>First Name:</label>
        <input type = "text" name="firstName" onChange={(e) => setFormData({...formData,
firstName: e.target.value})} /> <br />
        <label>Last Name:</label>
        <input type = "text" name="lastName" onChange={(e) => setFormData({...formData,
lastName: e.target.value})} /> <br />
        <label>Email:</label>
        <input type = "email" name="email" onChange={(e) => setFormData({...formData,
email: e.target.value})} /> <br />
        <label>Date of Birth:</label>
        <input type = "date" name="dob" onChange={(e) => setFormData({...formData, dob:
e.target.value})} /> <br />
        <button type="submit">Click Me!</button>
      </form>
      {submittedData && (
        <div style={{ marginTop: "20px", padding: "10px", border: "1px solid #ccc" }}>
          <h2>Submitted Data:</h2>
          <p>First Name: {submittedData.firstName}</p>
          <p>Last Name: {submittedData.lastName}</p>
          <p>Email: {submittedData.email}</p>
          <p>Date of Birth: {submittedData.dob}</p>
        </div>
      )}
    </div>
  )
}
export default UseForm
```

Form

Thank you for submitting the form!

First Name:

Last Name:

Email:

Date of Birth: ☐

Submitted Data:

First Name: hi

Last Name: hi

Email: hi@d

Date of Birth: 0001-01-01

DBMS

STRUCTURED DATA(SENSITIVE) : BANKING SERVICES
UNSTRUCTURED DATA : COMMENT SECTION

A DATABASE MANAGEMENT SYSTEM IS A SOFTWARE USED TO CREATE, STORE, MANAGE AND RETRIEVE DATA

STUDENT :

REG_NO	NAME	AGE
101	CCC	25
105	DDD	39
107	MMM	76

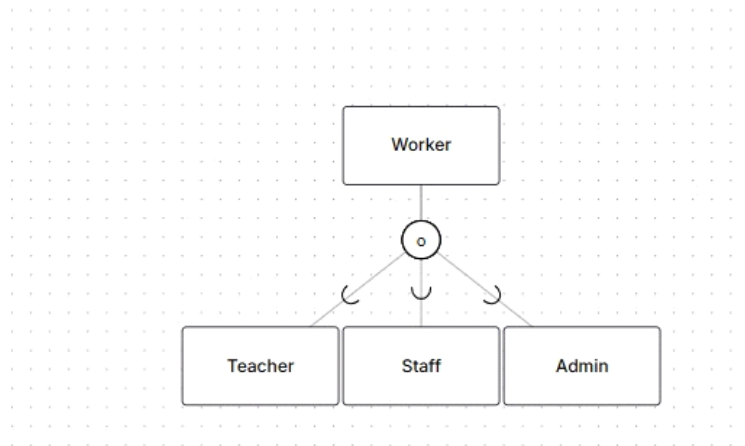
SCENARIO : TO MAINTAIN A DB FOR A SUPERMARKET, FIRSTLY IDENTIFY THE ITEMS PRESENT IN THE SUPERMARKET ALONG WITH THEIR QUALITIES LIKE PRICE, CONTENT AND TYPE. BEFORE DOING THAT, WE NEED TO DESIGN A DB MODEL. FOR THAT TO DEVELOP AN ALGORITHM.
DRAW THE ENTITY-RELATIONSHIP DIAGRAM WHICH ACTS AS A BLUEPRINT TO DESIGN THE DB. DRAW ER SEPARATELY FOR EACH TYPE OF THE ITEM AND JOIN ALL THE TYPES TO CREATE AN OPERATIONAL BLUEPRINT READY FOR DB.

NOTE : ERDPLUS IS A TOOL USED TO DRAW ER-DIAGRAM EFFUCKTIVELY.
AN ER DIAGRAM IS A VISUAL BLUEPRINT OF A DATABASE.

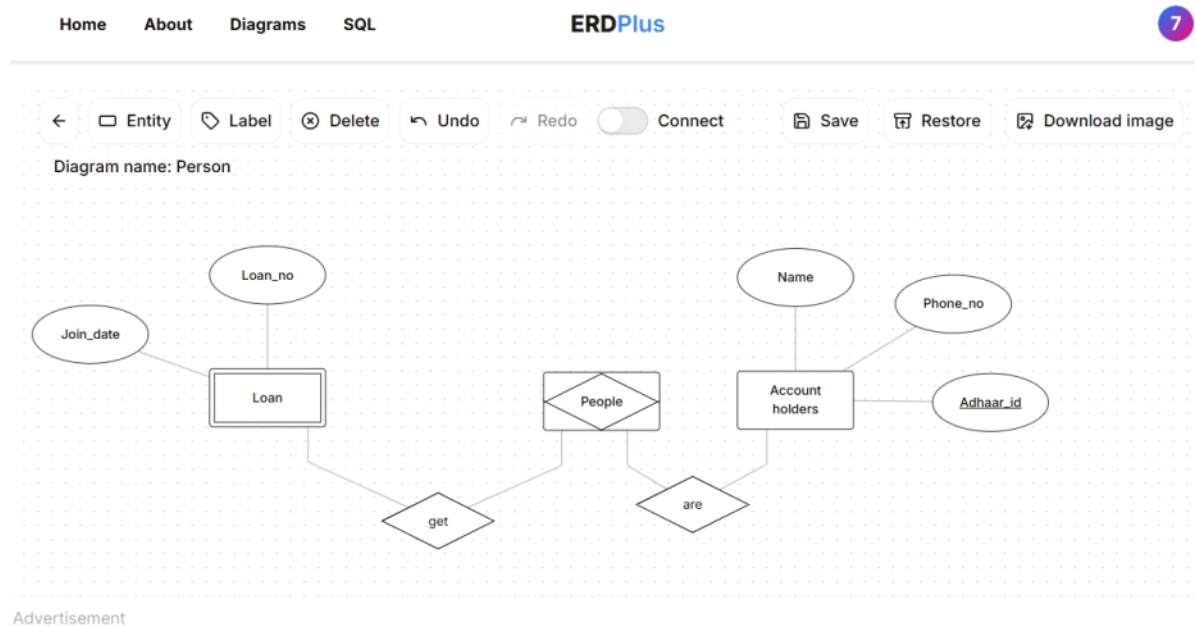
ER-DIAGRAM ESSENTIALS :

STRONG ENTITY	REGULAR RECTANGLE
WEAK ENTITY	DOUBLE RECTANGLE
ASSOSIATIVE ENTITY	SPECIFIES RELATIONSHIP. DIAMOND ENCLOSED WITHIN A RECTANGLE
SUPERTYPE ENTITY(DISJOINT)	CAN BE DISINTEGRATED INTO MULTIPLE ENTITIES EACH PERFORMING DIFFERENT ACTIVITIES. VEHICLES CAN BE CAR, BIKE, BUS.
SUPERTYPE ENTITY(OVERLAP)	WORKERS CAN BE STAFF, TEACHER, ADMIN.

DISJOINT:



AN EXAMPLE ER-DIAGRAM FOR PEOPLE TAKING LOANS IN BANKS:



TYPES OF ATTRIBUTES :

UNIQUE ATTRIBUTE	UNIQUE IDENTIFIER TO THE ENTITY. AKA THE PRIMARY KEY.	AN UNDERLINED ATTRIBUTE ENCLOSED WITHIN A CIRCLE.
COMPOSITE ATTRIBUTE	AN ATTRIBUTE CONTAINED FURTHER MORE ATTRIBUTES. NAME : FIRST, MIDDLE, LAST	AN ATTRIBUTE WITHIN A CIRCLE WHICH IS CONNECTED TO FURTHER MORE ATTRIBUTES OF THE SAME TYPE ENCLOSED WITHIN CIRCLES.
OPTIONAL ATTRIBUTES	EITHER CAN BE FILLED OR CAN BE LEFT OUT	A CIRCLE ATTRIBUTE FOLLOWED UP BY (O)
MULTIVALUED ATTRIBUTE	AN ATTRIBUTE CONTAINING MANY ATTRIBUTES OF SAME TYPE. A PERSON HAVING MANY PHONE NUMBERS	AN ATTRIBUTE WITHIN A CIRCLE WHICH IS CONNECTED TO FURTHER MORE ATTRIBUTES OF THE SAME TYPE ENCLOSED WITHIN CIRCLES.
DERIEVED ATTRIBUTE	AN ATTRIBUTE DERIVED FROM ANOTHER ATTRIBUTE. AGE DERIVED FROM DOB	ATTRIBUTES ARE ENCLOSED WITHIN DOTTED LINES OF CIRCLE.

TYPES OF CARDINALITIES :

ONE TO ONE :



ONE TO MANY(MANDATORY)



MANY

ONE TO ONLY ONE(MANDATORY)

ZERO TO ONE(OPTIONAL)

ZERO TO MANY(OPTIONAL)